

NØthing is Logical

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Slides: <https://www.marialoni.org/resources/SuB26.pdf>



Sinn und Bedeutung 31

Nantes Université, 4 September 2026

NØthing is logical (Nihil)

- **Goal of the project:** a formal account of a class of natural language inferences which deviate from classical logic
- **Common assumption:** these deviations are not logical mistakes, but consequence of pragmatic enrichments (Grice)
- **Initial strategy:** develop *logics of conversation* which model next to literal meanings also pragmatic factors and the additional inferences which arise from their interaction
- **Novel hypothesis:** **neglect-zero** tendency (a cognitive bias rather than a conversational principle) as crucial factor
- **Main conclusion:** deviations from classical logic consequence of enrichments albeit not (always) of the canonical Gricean kind
 - **Dominant view:** enrichments always product of **rational interactions** between cooperative language users
 - **Nihil hypothesis:** enrichments sometimes result from our preference to minimize cognitive effort (**bias-driven** vs product of rational choices)

Nihil team



Anttila



Degano



Klochowicz



Knudstorp



Ramotowska



Zhou



& more ...

Non-classical inferences

Free choice (FC)

- (1) FC: $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$ [von Wright 1968]
- (2) Deontic FC inference [Kamp 1973]
 - a. You may go to the beach *or* to the cinema.
 - b. $\rightsquigarrow$ You may go to the beach *and* you may go to the cinema.
- (3) Epistemic FC inference [Zimmermann 2000]
 - a. Mr. X might be in Victoria *or* in Brixton.
 - b. $\rightsquigarrow$ Mr. X might be in Victoria *and* he might be in Brixton.

Ignorance

- (4) The prize is either in the garden *or* in the attic $\rightsquigarrow$ The prize might be in the garden *and* might be in the attic [Grice 1989, p.45]
- (5) ? I have two *or* three children.
 - In the standard approach, **ignorance** is a conversational implicature
 - Less consensus on FC inferences analysed as conversational implicatures; grammatical (scalar) implicatures; semantic entailments; ...

The challenge of FC: adding FC to classical modal logic implies the equivalence of any two possibility claims

$$\Diamond a \Rightarrow_{\text{CML}} \Diamond(a \vee b) \Rightarrow_{\text{FC}} \Diamond b$$

Novel hypothesis: neglect-zero

- FC and ignorance inferences are [$\neq$ semantic entailments]
 - Not the result of Gricean reasoning [$\neq$ conversational implicatures]
 - Not the effect of applications of covert grammatical operators [$\neq$ grammatical (scalar) implicatures]
- They are rather a consequence of something else speakers do in conversation, namely,

NEGLECT-ZERO

when interpreting a sentence speakers construct models depicting reality (some verifying the sentence, some falsifying it) $\mapsto$ common assumption
and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*) $\mapsto$ novel hypothesis

- Tendency to neglect zero-models follows from the cognitive difficulty of:
 - ① conceiving emptiness, the absence of things rather than their presence
 - ② evaluating truths with respect to empty witness sets

[Nieder 2016; O. Bott et al 2019]

Novel hypothesis: neglect-zero

Illustration

- (6) Less than three squares are black.
- Verifier: [■, □, ■]
 - Falsifier: [■, ■, ■]
 - Zero-models: [□, □, □]; [■, ■, ■]; ...

Zero-models in (6-c) verify the sentence by virtue of an empty set of black squares

- Cognitive difficulty of zero-models confirmed by experimental findings and connected to / can be argued to explain:
 - the special status of 0 among the natural numbers [Nieder 2016]
 - why downward-monotonic quantifiers are more costly to process than upward-monotonic ones (*less* vs *more*) [O. Bott *et al* 2019]
- **N-Z hypothesis:** neglect-zero also at the origin of many common departures from classical reasoning
 - ① **FC and ignorance** [MA 2022]
 - ② **Existential Import:** every A is B $\Rightarrow$ some A is B
 - ③ **Principles from connexive logic:**
 - **Aristotle's Theses:** if not A then A $\Rightarrow \perp$; if A then not A $\Rightarrow \perp$
 - **Boethius' Thesis:** if A then B & if A then not B $\Rightarrow \perp$
[Ziegler, Knudstorp & MA 2026]

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

- | | |
|--|---|
| <p>(7) Maria ate an apple.</p> <p>a. Verifier: [ <p>b. Falsifiers: [ <p>c. Zero-models: none</p> </p></p> | <p>(8) Maria ate a banana.</p> <p>a. Verifier: [ <p>b. Falsifiers: [ <p>c. Zero-models: none</p> </p></p> |
| <p>(9) M ate an apple and a banana.</p> <p>a. Verifier: [ <p>b. Falsifiers: [ <p>c. Zero-models: none</p> </p></p> | <p>(10) M ate an apple or a banana.</p> <p>a. Verifier: ?</p> <p>b. Falsifiers: [ <p>c. Zero-models: ?</p> </p> |

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

- | | |
|--|---|
| <p>(11) Maria ate an apple.</p> <p>a. Verifier: [🍏]</p> <p>b. Falsifiers: [🍌]; [🍏]; []</p> <p>c. Zero-models: none</p> | <p>(12) Maria ate a banana.</p> <p>a. Verifier: [🍌]</p> <p>b. Falsifiers: [🍏]; [🍏]; []</p> <p>c. Zero-models: none</p> |
| <p>(13) M ate an apple and a banana.</p> <p>a. Verifier: [🍏🍌]</p> <p>b. Falsifiers: [🍏]; []</p> <p>c. Zero-models: none</p> | <p>(14) M ate an apple or a banana.</p> <p>a. Verifier: ?</p> <p>b. Falsifiers: [🍏]; []</p> <p>c. Zero-models: [🍏]; [🍌]</p> |
- Two **zero-models** in (14-c): verify the sentence by virtue of an empty witness for one of the disjuncts

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

- | | |
|---|--|
| <p>(15) Maria ate an apple.</p> <p>a. Verifier: [
b. Falsifiers: []; []; []
c. Zero-models: none</p> | <p>(16) Maria ate a banana.</p> <p>a. Verifier: [
b. Falsifiers: []; []; []
c. Zero-models: none</p> |
| <p>(17) M ate an apple and a banana.</p> <p>a. Verifier: [
b. Falsifiers: []; []
c. Zero-models: none</p> | <p>(18) M ate an apple or a banana.</p> <p>a. Verifier: [ ] $\Leftarrow$ 'split'
b. Falsifiers: []; []
c. Zero-models: []; []</p> |

- Two **zero-models** in (18-c): sentence verified by virtue of an empty witness for one of the disjuncts
- **Split state** in (18-a): multiple (possibly conflicting) alternatives processed in a parallel fashion
- **Neglect-zero hypothesis**: Zero-models, where only one of the disjuncts is depicted, are cognitively taxing and therefore disregarded
- The **exclusion of zero-models** from the set of verifiers for a disjunction leads directly to an account of ignorance inferences and FC

A new conjecture: no-split

(19) Maria ate an apple or a banana.

a. Verifier: [🍏 | 🍌]

b. Falsifiers: [🍏]; [🍌]

c. **Zero-models:** [🍏]; [🍌]

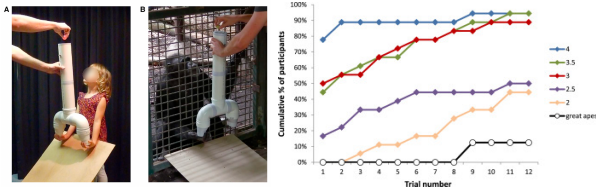
[⇐ **split state**]

- **Split states:** multiple (possibly conflicting) alternatives processed in a parallel fashion $\mapsto$ also a cognitively taxing operation

NO-SPLIT CONJECTURE

[Klochowicz, Sbardolini & MA, SuB 2025]

the ability to split states (entertain multiple possibilities) is developed late



Preschoolers have trouble conceiving possible different futures [Redshaw & Suddendorf 2016]

- **Proposal:** Combination of neglect-zero + no-split can explain non-classical interpretations of disjunction observed in pre-school children

No-split: conjunctive children

- Preschoolers sometimes (but systematically) interpret disjunctions conjunctively [Evans 93; Singh *et al* 16 (cf. Skordos *et al* 20); Cochard25; Bleotu *et al* 25]

(20) M ate an apple or a banana = M ate an apple and a banana
 $\alpha \vee \beta \equiv \alpha \wedge \beta$

(21) M can eat an apple or a banana = M can eat an apple and a banana
 $\Diamond(\alpha \vee \beta) \equiv \Diamond(\alpha \wedge \beta) \not\equiv \Diamond\alpha \wedge \Diamond\beta$

(22) M didn't eat an apple or a banana = M neither ate an apple nor a banana
 $\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$

- Proposal:** children have conjunctive readings as they (similarly to adults) neglect zero and, unlike adults, lack the ability to split

① Deriving ignorance:

Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍏 + 🍌 $\Rightarrow_{\text{SPLIT}}$ [🍏 | 🍌]
 $\leadsto$ It might be an apple and it might be a banana (adults)

② Deriving conjunctive reading:

Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍏 + 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [🍏 🍌]
 $\leadsto$ Both an apple and a banana (children)

③ In case of incompatible alternatives: [Leahy & Carey 2020, Bleotu *et al*]

Left OR right $\Rightarrow_{\text{NZ}}$ ✓ + ✗ $\Rightarrow_{\text{NO-SPLIT}}$ **contradiction** ($\perp$)
 $\leadsto$ Sequential interpretation (children)

No-split: homogeneity

- Ability to split a state not only required to interpret disjunction but also conjunction under negation
- Neglect-zero + no-split can also derive homogeneous interpretations [Sbardolini, 2023]

(23) M did *not* eat an apple *and* a banana $\leadsto$ M *neither* ate an apple *nor* a banana
 $\neg(\alpha \wedge \beta) \leadsto \neg\alpha \wedge \neg\beta$

- **New conjecture:** also adults tend to avoid split-states in cognitively taxing environments, notably under negation

① Deriving homogeneity:

NOT [apple AND banana] $\Rightarrow_{\text{NZ}}$ NO 🍏 + NO 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [NO 🍏 NO 🍌]
 $\leadsto$ Neither an apple nor a banana (children & no-splitting adults)

② Deriving ignorance:

NOT [apple AND banana] $\Rightarrow_{\text{NZ}}$ NO 🍏 + NO 🍌 $\Rightarrow_{\text{SPLIT}}$ [NO 🍏 | NO 🍌]
 $\leadsto$ Possibly not an apple and possibly not a banana (splitting adults)

- **Testable predictions arising from these conjectures:**
 - Increase of homogeneous (no-split) interpretations under cognitive load in adults (testable via dual-task studies)
 - Correlation between failure in Redshaw & Suddendorf's task and conjunctive behaviour in children

Cognitive bias approach (NIHIL⁺)

Common assumption: Reasoning and understanding of natural language involve the creation of mental models [Johnson-Laird 1983, Mascarenhas & Koralus 2013, a.o.]

- **Understanding** a sentence S means being able to mentally construct a model picturing the world which verifies S , and possibly also a model which falsifies it
- **Reasoning** depends on two main processes: first construct verifying models for the premises and then check the validity of the conclusion on these models

Novel hypothesis: biases can constrain the construction of these models and therefore impact both reasoning and interpretation:

- NEGLECT-ZERO prevents the constructions of zero-models;
- NO-SPLIT expresses a dispreference for split-states;
- ...

Cognitive bias approach (NIHIL⁺)

Comparison with competing accounts

	Disjunction ES-Quantifiers	Conditionals	Scalar implicature	Conjunctive or Homogeneity
Neo-Gricean/Relevance Grammatical view NIHIL ⁺	reasoning grammatical neglect-zero	reasoning — neglect-zero	reasoning grammatical [?]	— grammatical negl-z + no-split

NIHIL⁺: conversational inferences sometimes **bias-driven** (*fast, effortless*) rather than product of alternative-based reasoning (*slow, costly*)

Experimental evidence for neglect-zero

	REACTION TIME	ACQUISITION	PRIMING
Scalar implicature	delay	late	$SI_{some} \Rightarrow SI_{num}$
Neglect-zero effect	reverse delay	early	$NZ_{dis} \Rightarrow NZ_{esq}$
			$SI_{some} \not\Rightarrow NZ_{esq}$

[BKS⁺25], [DMR⁺25], [KSB⁺25], [RA25], [RKS⁺26]

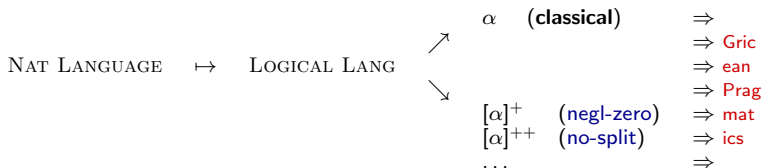
NEXT

- Modelling cognitive biases in team semantics
- Experimental study: ups and downs of ignorance [Degano *et al* 25]

Modelling biases in team semantics

General methodology

Natural language sentences translated into classical logic formulas interpreted in a **team semantics** which models both classical and enriched interpretations



Back to FC challenge

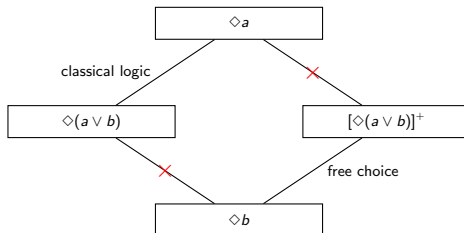


Figure: FC derived only for neglect-zero (NZ) enriched formulas

Modelling biases in team semantics

Team semantics

- Formulas interpreted wrt a set of points of evaluation (a **team**) rather than single ones [Hodges 1997; Väänänen 2007]

- Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

- Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features

- The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ **zero-models**
- Multi-membered teams can model parallel processing of alternatives $\mapsto$ **split states**

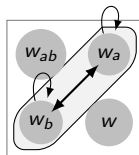


Figure: A multi-membered team modelling the parallel processing of possibilities *a* & *b*

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Neglect-zero & no-split

- Model-theoretic method**: biases modelled by disallowing empty (**neglect-zero**) and multi-membered teams (**no-split**)
- Syntactic method**: biases modelled via new logical atoms/operators
 - Neglect-zero**: via **non-emptiness atom** NE which disallows empty teams as possible verifiers [Yang & Väänänen 2017]

$$M, t \models \text{NE} \text{ iff } t \neq \emptyset$$
 - No-split**: via **flattening operator** F which induces pointwise evaluations and therefore mimics ban on parallel processing of alternatives

$$M, t \models F\phi \text{ iff for all } w \in t : M, \{w\} \models \phi$$

Modelling biases in team semantics

Classical and enriched interpretations

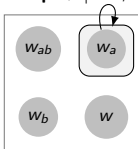
- $\alpha \Rightarrow$ empty team & split allowed $\mapsto$ **classical (literal) meaning**
- $[\alpha]^+ \Rightarrow$ empty team not allowed $\mapsto$ **NZ-enriched**
- $[\alpha]^{++} \Rightarrow$ neither empty team nor split allowed $\mapsto$ **NZ+NS-enriched**

Split disjunction in team semantics

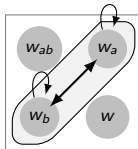
- Team t supports a **disjunction** iff t is the union of two subteams, each supporting one of the disjuncts
 - $a \vee b$: empty team allowed $\mapsto$ subteams can be empty **(classical)**
 - $[a \vee b]^+$: empty team not allowed $\mapsto$ subteams cannot be empty **(NZ-enr)**
 - $[a \vee b]^{++}$: neither empty team nor split allowed $\mapsto$ no empty or distinct subteams allowed **(NZ+NS-enriched)**

Empty Team Prop: $\emptyset \models \alpha$, for all classical α

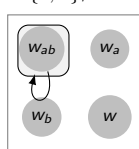
[PROP= $\{a, b\}$; $W = \{w_{ab}, w_a, w_b, w\}$ **]**



$\models a \vee b$
 $\not\models [a \vee b]^+$
 $\not\models [a \vee b]^{++}$
 $\mapsto$ **Zero-model**



$\models a \vee b$
 $\models [a \vee b]^+$
 $\not\models [a \vee b]^{++}$
 $\mapsto$ **No-zero, split**



$\models a \vee b$
 $\models [a \vee b]^+$
 $\models [a \vee b]^{++}$
 $\mapsto$ **No-zero, no-split**

Modelling biases in team semantics

(24) Maria ate an apple or a banana.

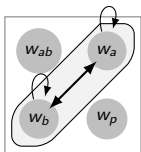
$[a \vee b]$

a. **No-zero, split** verifier: [🍏 | 🍌]

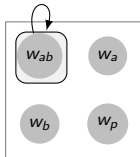
b. **No-zero, no-split** verifier: [🍏 🍌]

c. **Zero-models**: [🍏] ; [🍌]

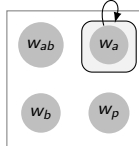
d. **Falsifier**: [🍏]



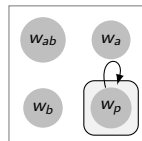
No-zero, split verifier



No-zero, no-split verifier



Zero-model



Falsifier

Predictions

We derive **ignorance** & **conjunctive** interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(25) M ate an apple or a banana $\rightsquigarrow_{nz}$ It might be an apple and it might be a banana

$$[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta \quad (\text{where } R \text{ is state-based})$$

(26) M ate an apple or a banana $\rightsquigarrow_{ns}$ M ate both apple and banana

$$[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$$

(27) M didn't eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M ate neither apple nor banana

$$[\neg(\alpha \vee \beta)]^{(++)} \models \neg\alpha \wedge \neg\beta \quad \text{and more predictions including FC, homogeneity, ...}$$

A closer look at these predictions

Many enrichments triggered by disjunction

- (28) Maria ate an apple or a banana ($\alpha \vee \beta$) $\leadsto$
- a. **Ignorance**: speaker doesn't know which
 - b. **Conjunctive interpretation**: both $(\alpha \wedge \beta)$
 - c. **Exclusive interpretation**: not both $\neg(\alpha \wedge \beta)$

Can we capture all these enrichments as nz- or ns-effects?

Two components of full ignorance: possibility vs uncertainty

- (29) Maria ate an apple or a banana $\leadsto$ speaker doesn't know which
[Degano *et al* 2025]¹
- a. **Possibility**: It is possible that M ate an apple and it is possible that M ate a banana $\Diamond_e \alpha \wedge \Diamond_e \beta$
 - b. **Uncertainty**: It is uncertain that M ate an apple and it is uncertain that M ate a banana $\neg \Box_e \alpha \wedge \neg \Box_e \beta$

¹Degano, Marty, Ramotowska, MA, Sudo, Romoli, Breheny. The ups and downs of ignorance. *Natural Language Semantics* (2025)

Neglect-zero effects on disjunction: predictions of []⁺...-enrichment

Many no-zero verifiers for nz-enriched disjunction

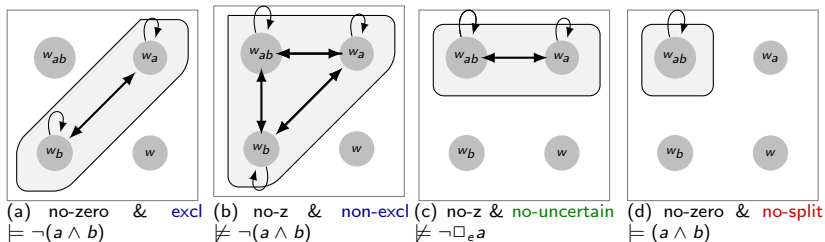


Figure: Models for enriched $[a \vee b]^+$.

- 1 Neglect-zero enrichment derives **possibility**: $[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$
- 2 Neglect-zero + no-split derives **conjunctive or**: $[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$
- 3 Neglect-zero enrichment does not derive **exclusive readings**:
 $[\alpha \vee \beta]^+ \not\models \neg(a \wedge b)$
- 4 Neglect-zero enrichment does not derive **uncertain inferences**:
 $[\alpha \vee \beta]^+ \not\models \neg \Box_e a$
 $\mapsto$ in contrast to standard neo-Gricean approach to ignorance



Two derivations of full ignorance

① Standard neo-Gricean derivation

[Sauerland 2004]

(i) Uncertainty derived through **quantity** reasoning

(30) $\alpha \vee \beta$ ASSERTION

(31) $\neg \Box_e \alpha \wedge \neg \Box_e \beta$ UNCERTAINTY (from QUANTITY)

(ii) Possibility derived from uncertainty and **quality** about assertion

(32) $\Box_e(\alpha \vee \beta)$ QUALITY ABOUT ASSERTION

(33) $\Rightarrow \Diamond_e \alpha \wedge \Diamond_e \beta$ POSSIBILITY

② Neglect-zero derivation

(i) Possibility derived as **neglect-zero** effect

(34) $\alpha \vee \beta$ ASSERTION

(35) $\Diamond_e \alpha \wedge \Diamond_e \beta$ POSSIBILITY (from NEGLECT-ZERO)

(ii) Uncertainty derived from possibility and **scalar reasoning**

(36) $\neg(\alpha \wedge \beta)$ SCALAR IMPLICATURE

(37) $\Rightarrow \neg \Box_e \alpha \wedge \neg \Box_e \beta$ UNCERTAINTY

The ups and downs of ignorance [Degano et al 2025]

Contrasting predictions of competing accounts of ignorance

- **Neo-Gricean**: No possibility without uncertainty
- **Neglect-zero**: Possibility derived independently from uncertainty

Experimental findings

- Using adapted mystery box paradigm, compared conditions in which
 - both uncertainty and possibility are false [zero-model]
 - uncertainty false but possibility true [no-zero, no-uncertain model]
- **Zero-models** far more rejected (66%) than **no-uncertain model** (5%)
 ⇒ Evidence that possibility can arise without uncertainty
- A challenge for the traditional neo-Gricean approach

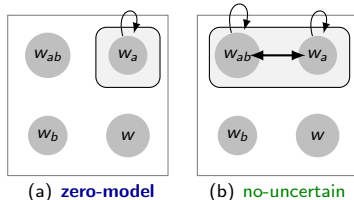


Figure: Models for $(a \vee b)$

Conclusions

- **FC, ignorance inference:** a mismatch between logic and language
- **Grice's insight:**
 - stronger meanings can be derived paying more “attention to the nature and importance to the conditions governing conversation”
- **Nihil hypothesis:** some strengthenings due to cognitive tendencies rather than rational choices
 - FC, possibility and related inferences as neglect-zero effects

Literal meanings (classical fragment) + cognitive factor (NZ) $\Rightarrow$ FC, possibility, connexive principles, etc
 - Conjunctive *or*, homogeneity as no-zero + no-split effect

Literal meanings (classical fragment) + cognitive factors (NZ, NS) $\Rightarrow$ conjunctive *or*, homogeneity
- Implementation in team-based modal logics
- **Ups and downs of ignorance:** a challenge for the neo-Gricean approach

THANK YOU!²

²This work is supported by NWO OC project *Nothing is Logical* (grant no 406.21.CTW.023).

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BSML: Classical ML + NE

[focus on neglect-zero modelled syntactically]

Language

$$\phi := p \mid \neg\phi \mid \phi \vee \phi \mid \phi \wedge \phi \mid \Diamond\phi \mid \text{NE}$$

Bilateral team semantics

Given Kripke model $M = \langle W, R, V \rangle$ & teams/states $s, t, t' \subseteq W$

$$M, s \models p \quad \text{iff} \quad \text{for all } w \in s : V(w, p) = 1$$

$$M, s \models\!\!\!\models p \quad \text{iff} \quad \text{for all } w \in s : V(w, p) = 0$$

$$M, s \models \neg\phi \quad \text{iff} \quad M, s \models\!\!\!\models \phi$$

$$M, s \models\!\!\!\models \neg\phi \quad \text{iff} \quad M, s \models \phi$$

$$M, s \models \phi \vee \psi \quad \text{iff} \quad \text{there are } t, t' : t \cup t' = s \text{ \& } M, t \models \phi \text{ \& } M, t' \models \psi \quad \Leftarrow$$

$$M, s \models\!\!\!\models \phi \vee \psi \quad \text{iff} \quad M, s \models\!\!\!\models \phi \text{ \& } M, s \models\!\!\!\models \psi$$

$$M, s \models \phi \wedge \psi \quad \text{iff} \quad M, s \models \phi \text{ \& } M, s \models \psi$$

$$M, s \models\!\!\!\models \phi \wedge \psi \quad \text{iff} \quad \text{there are } t, t' : t \cup t' = s \text{ \& } M, t \models\!\!\!\models \phi \text{ \& } M, t' \models\!\!\!\models \psi$$

$$M, s \models \Diamond\phi \quad \text{iff} \quad \text{for all } w \in s : \exists t \subseteq R[w] : t \neq \emptyset \text{ \& } M, t \models \phi$$

$$M, s \models\!\!\!\models \Diamond\phi \quad \text{iff} \quad \text{for all } w \in s : M, R[w] \models\!\!\!\models \phi \quad [\text{where } R[w] = \{v \in W \mid wRv\}]$$

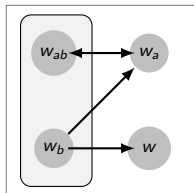
$$M, s \models \text{NE} \quad \text{iff} \quad s \neq \emptyset$$

$$M, s \models\!\!\!\models \text{NE} \quad \text{iff} \quad s = \emptyset \quad [\text{Empty Team Prop: } \emptyset \models \alpha, \text{ for all NE-free } \alpha]$$

Entailment: $\phi_1, \dots, \phi_n \models \psi$ iff for all M, s : $M, s \models \phi_1, \dots, M, s \models \phi_n \Rightarrow M, s \models \psi$

Proof Theory: MA, Anttila & Yang (2024); **Expressive completeness:** Anttila & Knudstorp (2025);

Comparisons via translation into Modal Information Logic: Knudstorp et al (2026)



Neglect-zero effects in BSML: neglect-zero enrichment

Neglect-zero modelled syntactically using NE

Non-emptiness

NE is supported in a state if and only if the state is not empty

$$M, s \models_{\text{NE}} \quad \text{iff} \quad s \neq \emptyset$$

$$M, s \not\models_{\text{NE}} \quad \text{iff} \quad s = \emptyset$$

Neglect-zero enrichment function

For NE-free α , $[\alpha]^+$ defined as follows:

$$[p]^+ = p \wedge \text{NE}$$

$$[\neg\alpha]^+ = \neg[\alpha]^+ \wedge \text{NE}$$

$$[\alpha \vee \beta]^+ = ([\alpha]^+ \vee [\beta]^+) \wedge \text{NE}$$

$$[\alpha \wedge \beta]^+ = ([\alpha]^+ \wedge [\beta]^+) \wedge \text{NE}$$

$$[\Diamond\alpha]^+ = \Diamond[\alpha]^+ \wedge \text{NE}$$

$[]^+$ enriches formulas with the requirement to satisfy NE distributed along each of their subformulas

Neglect-zero effects on quantifiers

Predictions of qBSML[→] (with max info)

- (38) Less than three squares are black $\mapsto \forall xyz((Sx \wedge Bx \wedge \dots) \rightarrow (x = y \vee \dots))$
- a. Verifier: [■, □, ■]
 - b. Falsifier: [■, ■, ■]
 - c. Zero-models: [□, □, □]; [▲, ▲, ▲]; ... $\leadsto_{nz}$ there are black squares
- (39) Every square is black. $\mapsto \forall x(Sx \rightarrow Bx)$
- a. Verifier: [■, ■, ■]
 - b. Falsifier: [■, □, ■]
 - c. Zero-models: [△, △, △]; [▲, ▲, ▲]; ... $\leadsto_{nz}$ there are squares
- (40) No squares are black. $\mapsto$ (i) $\forall x(Sx \rightarrow \neg Bx)$; (ii) $\neg \exists x(Sx \wedge Bx)$
- a. Verifier: [□, □, □]
 - b. Falsifier: [■, □, □]
 - c. Zero-models for (i): [△, △, △]; [▲, ▲, ▲]; ... $\leadsto_{nz}$ there are squares
 - d. Zero-models for (ii): none no neglect-zero effect
- (41) Every square is red or white. $\mapsto \forall x(Sx \rightarrow (Rx \vee Wx))$
- a. Verifier: [■, □, ■]
 - b. Falsifier: [■, □, ■]
 - c. Zero-models: [■, ■, ■]; [□, □, □] $\leadsto_{nz}$ there are white sqrs & red sqrs